

Complex Analysis – 5

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Теоретический минимум.

Опр. $\text{Aut}(\mathbb{D})$ is the group of biholomorphic maps of the unit disc onto itself; the *Blaschke factor* is $b_a(z) = \frac{z-a}{1-\bar{a}z}$, and a *finite Blaschke product* is $e^{i\theta} \prod_{k=1}^n b_{a_k}(z)$.

Теорема 1 (лемма Шварца) *If $f: \mathbb{D} \rightarrow \mathbb{D}$ is holomorphic and $f(0) = 0$, then $|f(z)| \leq |z|$ and $|f'(0)| \leq 1$; equality at one interior point forces $f(z) = e^{i\theta}z$.*

Теорема 2 (Шварц–Пик) *For any holomorphic $f: \mathbb{D} \rightarrow \mathbb{D}$,*

$$\frac{|f'(z)|}{1-|f(z)|^2} \leq \frac{1}{1-|z|^2}, \quad \left| \frac{f(z)-f(w)}{1-\overline{f(w)}f(z)} \right| \leq \left| \frac{z-w}{1-\bar{w}z} \right|,$$

with equality at one point if and only if $f \in \text{Aut}(\mathbb{D})$; i.e. holomorphic self-maps of the disc do not increase the hyperbolic metric $\frac{2|dz|}{1-|z|^2}$.

Факт 1 $\text{Aut}(\mathbb{D}) = \{e^{i\theta}b_a : \theta \in \mathbb{R}, a \in \mathbb{D}\}$; every automorphism is determined by the image of one point together with the argument of the derivative there.

- (S.T. Yau CSMC 2021) Determine $\text{Aut}(\mathbb{D})$, the group of holomorphic automorphisms of the unit disc, and describe the subgroup fixing the origin.
- (S.T. Yau CSMC 2013, team) Prove that for every holomorphic $f: \mathbb{D} \rightarrow \mathbb{D}$

$$\frac{|f'(z)|}{1-|f(z)|^2} \leq \frac{1}{1-|z|^2}, \quad z \in \mathbb{D},$$

and that equality at a single point z_0 forces $f \in \text{Aut}(\mathbb{D})$ and equality everywhere.

3. (S.T. Yau CSMC 2016) Suppose f is analytic on $\mathbb{D}$ with $|f(z)| \leq 1$, and $f(\frac{1}{2}) = f(-\frac{1}{2}) = 0$. Prove that $|f(0)| \leq \frac{1}{4}$.
4. (S.T. Yau CSMC 2017, team) Let f be holomorphic on $\mathbb{D}$ with $|f| < 1$, and let $\alpha \neq \beta$ be points of $\mathbb{D}$ with $f(\alpha) = f(\beta) = 0$. Prove that

$$|f(z)| \leq \left| \frac{z - \alpha}{1 - \bar{\alpha}z} \cdot \frac{z - \beta}{1 - \bar{\beta}z} \right|, \quad z \in \mathbb{D}.$$

5. (S.T. Yau CSMC 2014) Let $f_1, \dots, f_n$ be analytic on $\mathbb{D}$ and continuous on $\bar{\mathbb{D}}$. Prove that $\varphi(z) = |f_1(z)| + \dots + |f_n(z)|$ attains its maximum on the boundary $\partial\mathbb{D}$.